

Use of Floquet Buffer

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Part I

Sparsho: Use of Floquet Buffer (sublayer) in Thermodynamic neuron

1 Introduction

1.1 Current Problem in the Thermodynamic neuron architecture

In open-quantum systems we usually consider an effective picture to study a system as because the total entropy production rate of the system along with the bath is zero meaning:

$$\partial_t S_{vn}[\hat{\rho}_{tot}] = 0$$

so in order to define a non trivial entropy rate production which is done by defining an effective density matrix:

$$\hat{\rho}_{effective} = \hat{\rho}_s(t) \otimes_{\alpha} \hat{\tau}_{\beta_{\alpha}, \mu_{\alpha}}$$

where $\hat{\rho}_s(t)$ is essentially the pure states or the sum of the pure states of the total density matrix as $\hat{\rho}_{total}(t)$ meaning we can write this as :

$$\hat{\rho}_s(t) = -Tr\{\hat{\rho}_{tot}(t)\}$$

this is so-called the reduced density matrix

As we are considering a relative description the information lost is essentially given by then the quantum relative entropy is given by:

$$S[\hat{\rho}_{tot}][\hat{\rho}_{effective}] = S_{vn}[\hat{\rho}_s] + \sum_{\alpha} \beta_{\alpha} \langle \hat{H}_{\alpha} - \mu_{\alpha} \hat{N}_{\alpha} \rangle + \sum_{\alpha} \ln Z_{\alpha} - S_{vn}[\hat{\rho}_{tot}]$$

now if we differentiate both sides with time we essentially consider that the total entropy change of the system is zero and the partition function of the bath is also invariant in time so effectively we get that :

$$\Gamma = \partial_t S_{vn}[\hat{\rho}_s] + \frac{1}{k_{\beta}} \sum_{\alpha} \frac{J_{\alpha}}{T_{\alpha}}$$

this entropy is not ensured to be positive in case of non markovian systems as there might be a backflow of information from the bath. In the current architectures of thermodynamics based logic gates all of them work as autonomous machines meaning they do not need external clock pulses for logical output. So the issue here is that we can't control pretty much any of these except for the intrinsic properties of the system. In modern CMOS based gates operate by maintaining high fidelity as because its terminals are isolated from each other intrinsically but that is not possible for qubit. Due to these disadvantages the thermodynamic not gate being analogous to perceptron is an interesting architecture but it here the question arises is there any way to develop robust thermodynamic gates without this information backflow or as the system exchanges information with the environment is there a way to feedback lost information for increasing accuracy?

1.2 Proposed Floquet Buffer architecture

In the original Markovian thermodynamic neuron, the computational subsystem is directly coupled to four thermal reservoirs that act as the input, output, reference, and modulator baths. Since the qubits are in continuous contact with these reservoirs, information transfer is not strictly unidirectional. Energy and information may flow from the baths to the qubits as well as from the qubits back into the baths. Consequently, part of the logical information encoded within the neuron can be lost to the environment before reaching the output channel. Furthermore, the logical state is inferred from observables associated with the output bath, making the readout process susceptible to thermal fluctuations, environmental noise, and measurement errors.

To overcome these limitations, we propose the insertion of periodically driven Floquet buffer layers between the neuron and selected reservoirs. Specifically, a Floquet buffer is introduced between the reference qubit and the reference bath, while another Floquet buffer is placed in the output branch between the collector subsystem and the output bath. The resulting architecture introduces an intermediate dynamical layer between the computational subsystem and its environment.

Rather than a direct coupling

$$\mathcal{C} \longleftrightarrow \mathcal{B},$$

where $\mathcal{C}$ denotes the computational subsystem and $\mathcal{B}$ denotes a thermal reservoir, the interaction is mediated through a periodically driven Floquet subsystem $\mathcal{F}$,

$$\mathcal{C} \longleftrightarrow \mathcal{F} \longleftrightarrow \mathcal{B}.$$

Here, $\mathcal{F}$ acts as an intermediate Floquet buffer whose Hamiltonian satisfies

$$H_F(t + T) = H_F(t),$$

with driving period $T = 2\pi/\Omega$. The periodic modulation generates a quasienergy structure that selectively controls the transfer of energy and information between the computational subsystem and the reservoir.

From a physical perspective, the Floquet buffer acts as a spectral and temporal filter. By engineering the driving frequency and modulation strength, unwanted transitions can be suppressed while desired energy-transfer channels are enhanced. As a result, the effective system-bath coupling becomes dynamically controllable. The Floquet layer therefore reduces information leakage into the reservoirs and simultaneously suppresses noise entering the computational subsystem from the environment.

In the output branch, the Floquet buffer can additionally implement a feedforward mechanism. Information extracted from the collector subsystem is first processed by the driven Floquet layer before reaching the output reservoir. The output signal therefore experiences an effective amplification and stabilization stage, improving the distinguishability of logical states and reducing readout errors.

Conceptually, this architecture is analogous to a CMOS voltage inverter. In a CMOS circuit, transistor networks provide isolation between the input and output while simultaneously enabling gain, noise suppression, and controlled signal propagation. Similarly, the Floquet buffers provide dynamical isolation between the thermodynamic neuron and its thermal environment while actively regulating energy transport through periodic driving. The periodically driven Floquet layers therefore function as quantum thermodynamic counterparts of the intermediate transistor stages found in conventional CMOS logic circuits. We have done a mathematical rigorous note in Part [II](#).

2 What is a Floquet Subsystem?

Suppose there is Hamiltonian which satisfies this relation:

$$H(t + T) = H(t)$$

where T is the time period and is given by

$$T = \frac{2\pi}{\omega}$$

Such periodically driven systems are called Floquet Systems. Now the Floquet theory is the time analogous version of the Bloch theory which states that the solution of a Schrödinger equation given by:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

can be written as

$$|\psi_\alpha(t)\rangle = e^{-i\epsilon_\alpha t/\hbar} |\phi_\alpha(t)\rangle$$

where ϵ_α is the quasi energies.

2.1 Practical Examples of Floquet Engineering

A familiar example of floquet engineering arises in frequency modulation (FM) communication systems. In FM transmission, the frequency of a high-frequency carrier signal is varied periodically according to an information-bearing signal. Mathematically, an FM signal can be expressed as

$$s(t) = A_c \cos [\omega_c t + \beta \sin(\omega_m t)]$$

where A_c is the carrier amplitude, ω_c is the carrier frequency, ω_m is the modulation frequency, and β is the modulation index. The sinusoidal term periodically modulates the phase and consequently the instantaneous frequency of the carrier signal.

For example, in FM radio broadcasting, an audio signal periodically modulates the carrier frequency, allowing information to be encoded and transmitted efficiently. This provides an intuitive analogy to Floquet engineering, where a physical system is subjected to a periodic drive. Just as periodic modulation in FM generates new spectral components around the carrier frequency, periodic driving in a Floquet system can generate new dynamical features and effective energy structures that are absent in the undriven system.

3 Floquet Buffer Extension to the Thermodynamic Neuron

Before creating any extension and complex architecture to the thermodynamic neuron we must first do a mathematical and numerical rigorous note on the original architecture and then create viable extensions from it.

3.1 Original Thermodynamic Neuron architecture

In the thermodynamic neuron gate architecture there are 2 parts:

- **Collector:** This is consists of 3 interacting qubits C_0, C_1, C_Z
- **Modulator:** This consists of a single Qubit C_M connecting the collector , through another bath B_r

As shown in the Fig: 1 you can see that the collector module has 3 qubits amongsts which 2 qubits C_0, C_1 are connected to baths B_0, B_1 amongst which the bath B_0 is the reference bath meaning having a constant temperature. The B_1 is the bath which through which the information is encoded as input and passes through the collector. The 2 qubits are connected and directly interacting with the 3rd qubit C_z which is connected to a finite bath B_z which is finite reservoir with heat capacity $\mathcal{C}$. We essentially take the logical output as function of this bath in steady state. One might ask why is the modulator needed when the logical output is measured from B_z^∞ !!

Essentially without the modulator the thermodynamic not gate essentially acts as linear amplifier not a robust logical gate. The ultimate goal of the collector is to create something called virtual temperature which we will discuss later in the note and it tries to equilibrate the logical output to the virtual temperature making the input directly to the output i.e:

$$B_Z^\infty \approx B_v$$

with the modulator is essentially its job is to compete with the collector and constrain the possible values of B_Z^∞ acting as a filter to give the logical output.

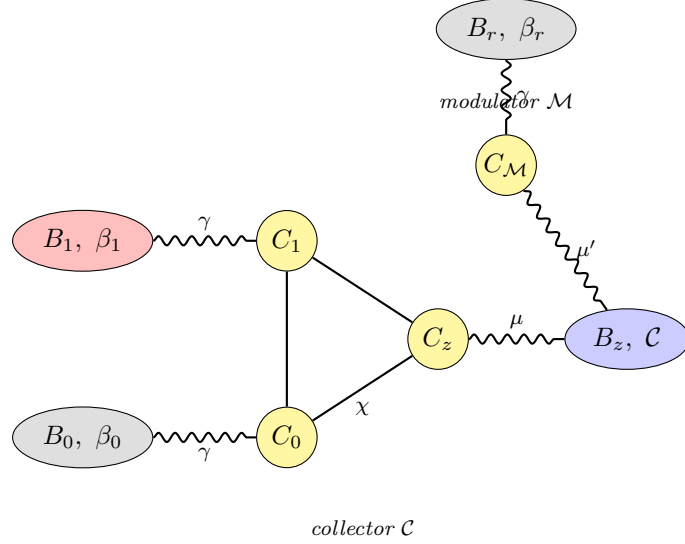


Figure 1: Architecture of the original (autonomous) thermodynamic neuron. Yellow: machine qubits; red: input bath encoding β_1 ; grey: fixed reference baths β_0, β_r ; blue: finite output reservoir B_z (heat capacity $\mathcal{C}$) from which the logical output is read. Wavy lines are dissipative (thermal) contacts with reset rates γ, μ, μ' ; solid lines depict the single energy-preserving three-body interaction H_{int} [Eq. (3), drawn as the C_0 – C_1 – C_z triangle], not three independent couplings. The collector builds the virtual temperature β_v ; the modulator $C_{\mathcal{M}}$, straddling B_r and B_z , saturates the response.

3.1.1 The complete Hamiltonian of the machine, term by term

The overview above, and the entire population / virtual-temperature analysis that follows, are consequences of a single microscopic object: the total Hamiltonian of the machine together with its reservoirs. Because the rest of Part 1 works with populations and reset-model rates, it is easy to forget that a Hamiltonian is generating them. This subsection writes *every* term explicitly, states the Hilbert space it acts on, and — most importantly — justifies why each term has the form it does and why the terms one might naively expect are deliberately absent. The reset rates γ, μ, μ' used from Sec. 3.1.3 onward are nothing but the second-order (Fermi golden-rule) reductions of the couplings written here; the population picture is the weak-coupling shadow of this Hamiltonian.

The total Hamiltonian. The machine lives on $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_{\mathcal{M}} \otimes \bigotimes_{\alpha} \mathcal{H}_{B_{\alpha}}$, where $\mathcal{H}_S = (\mathbb{C}^2)^{\otimes 3}$ carries the three collector qubits C_0, C_1, C_z , $\mathcal{H}_{\mathcal{M}} = \mathbb{C}^2$ the modulator qubit $C_{\mathcal{M}}$, and $\mathcal{H}_{B_{\alpha}}$ the reservoirs $\alpha \in \{0, 1, z, r\}$. The generator of the dynamics is

$$H_{\text{tot}}(t) = \underbrace{H_S + H_{\text{int}}}_{\text{collector}} + \underbrace{H_{\mathcal{M}}}_{\text{modulator}} + \underbrace{\sum_{\alpha} H_{B_{\alpha}}}_{\text{reservoirs}} + \underbrace{\sum_{\alpha} H_{SB_{\alpha}}}_{\text{system-bath}} + \underbrace{\Delta H_F(t)}_{\text{Floquet extension (Sec. 3.2)}}. \quad (1)$$

The last term is zero for the original (autonomous) neuron and is switched on by the buffer insertion. We take each block in turn.

(1) Free qubit Hamiltonian H_S : why two levels, why diagonal.

$$H_S = \epsilon_0 n_0 + \epsilon_1 n_1 + \epsilon_z n_z, \quad n_i := |1\rangle\langle 1|_i = \sigma_+^{(i)} \sigma_-^{(i)} = \frac{1}{2}(\mathbb{I} - \sigma_z^{(i)}). \quad (2)$$

- *Why qubits.* Each node is the two lowest levels of a physical mode with a well-separated gap ϵ_i ; at the temperatures used, higher levels are thermally inaccessible, so the truncation to $\{|0\rangle, |1\rangle\}$ is controlled.

- *Why only σ_z (no bare transverse field).* The logical variable is a *population*, i.e. a thermal occupation read in the energy eigenbasis. H_S is therefore chosen diagonal in the computational basis so that the readout basis is an eigenbasis: a bare σ_x term would rotate the eigenbasis away from the measurement basis and coherently mix the two logical values. The bare qubits must be “idle”—their sole job is to hold populations that the baths and H_{int} set.
- *Why the constants are dropped.* $n_i = \frac{1}{2}(\mu - \sigma_z^{(i)})$ differs from $\frac{1}{2}\sigma_z^{(i)}$ only by an additive constant, which shifts the zero of energy and never enters any observable.
- *Resonance conditions (engineered, not automatic).* We impose $\epsilon_z = \epsilon_0 - \epsilon_1$ (so C_z is resonant with the virtual qubit, below) and $\epsilon_{\mathcal{M}} = \epsilon_z$ (so the modulator is resonant with the output channel). These are hardware design constraints on the fabricated gaps.

(2) The collector interaction H_{int} : the term that does the computation. This is the term the population analysis presupposes but never writes. With the ordering $(C_0 C_1 C_z)$,

$$H_{\text{int}} = g(|100\rangle\langle 011| + |011\rangle\langle 100|) = g(\sigma_+^{(0)}\sigma_-^{(1)}\sigma_-^{(z)} + \sigma_-^{(0)}\sigma_+^{(1)}\sigma_+^{(z)}). \quad (3)$$

Five points justify its form.

- (i) **Energy conservation / RWA.** The state $|100\rangle$ has energy ϵ_0 and $|011\rangle$ has $\epsilon_1 + \epsilon_z$; under the resonance $\epsilon_z = \epsilon_0 - \epsilon_1$ these are *degenerate*. Starting from a generic three-qubit exchange coupling and passing to the interaction picture, every product of $\sigma_{\pm}$ ’s rotates at a signed sum of the ϵ_i ; since $g \ll \epsilon_i$, all of them average to zero over the free evolution *except* the resonant pair, which is stationary. That surviving, secular piece is exactly (3).
- (ii) **Why three-body, not pairwise.** The virtual qubit $\{|01\rangle, |10\rangle\}$ is a *joint* property of C_0 and C_1 . A two-body term $\sigma_+^{(0)}\sigma_-^{(1)}$ would only hop the excitation between C_0 and C_1 (an internal rotation of the virtual qubit) and transfer nothing to C_z . Reading the virtual qubit with C_z therefore requires flipping all three spins at once—an irreducibly three-body operator.
- (iii) **The swap interpretation.** In virtual-qubit variables $|1\rangle_v = |10\rangle$, $|0\rangle_v = |01\rangle$, Eq. (3) reads $H_{\text{int}} = g(|1\rangle_v\langle 0|_z\langle 0|_v\langle 1|_z + \text{h.c.})$: a resonant SWAP (beam splitter) between the virtual qubit and the target C_z . *This is the microscopic reason C_z equilibrates to the virtual temperature β_v* —the interaction literally swaps the virtual qubit’s population into C_z . The steady-state population argument of Sec. 3.1.2 is the fixed point of this swap.
- (iv) **Why the non-resonant partners are absent.** Terms such as $\sigma_+^{(0)}\sigma_+^{(1)}\sigma_+^{(z)}$ create three excitations and rotate at $\epsilon_0 + \epsilon_1 + \epsilon_z$; they would inject energy with no bath to supply it. The RWA discards them with relative error $\mathcal{O}(g/\epsilon)$.
- (v) **Why no bare C_0 – C_1 or $\sigma_z\sigma_z$ coupling.** The three lines drawn among C_0, C_1, C_z in Fig. 2 depict the single three-body term (3), *not* three independent pairwise couplings. A genuine C_0 – C_1 exchange or a dispersive $\sigma_z^{(0)}\sigma_z^{(1)}$ term would hybridise the two input qubits and spoil the product-thermal structure $\tau_0 \otimes \tau_1$ from which the virtual temperature β_v is *defined*; it is deliberately omitted.

Remark 3.1. The resonant pair is $|100\rangle \leftrightarrow |011\rangle$. A tempting but *wrong* choice is $|101\rangle \leftrightarrow |010\rangle$, which is detuned by $\epsilon_0 + \epsilon_z - \epsilon_1 = 2\epsilon_z$ and, at $g \ll \epsilon_z$, is dynamically inert—the collector would not function. The correct pair is fixed unambiguously by $\epsilon_0 = \epsilon_1 + \epsilon_z$.

(3) Why a bath at all, and the reservoir Hamiltonians. Before writing H_{B_α} we answer the conceptual question: why couple to a bath?

- *A closed system cannot compute by relaxation.* Unitary evolution is reversible, has no attractor, and keeps $S_{vn}[\hat{\rho}_{\text{tot}}]$ constant; there is no “settling into an answer.” The baths supply the dissipation that drives the machine to a *unique* nonequilibrium steady state—the logical output. This is why the effective entropy production $\Gamma = \partial_t S_{vn}[\hat{\rho}_s] + \frac{1}{k_B} \sum_\alpha J_\alpha/T_\alpha$ of Sec. 1.1 is nonzero only when the baths are present.
- *The baths **are** the input register.* The logical input β_1 is a bath temperature; the bias β_0 is a bath temperature. There is nowhere else in the model to encode the inputs.
- *The temperature gradient is the power supply.* The autonomous (clockless) gate is driven entirely by heat flowing from hot to cold reservoirs; with no gradient there is no current and no logic.

A bath is thus simultaneously the input register, the power supply, and the source of irreversibility that makes readout possible—not an optional add-on. Each reservoir is the standard bosonic (Caldeira–Leggett) continuum,

$$H_{B_\alpha} = \sum_k \omega_{\alpha k} a_{\alpha k}^\dagger a_{\alpha k}, \quad (4)$$

a dense harmonic spectrum whose continuum limit yields genuine irreversible decay (Poincaré recurrence sent to infinity) and whose Gaussian statistics make it the exact leading-order model of any weakly coupled environment with the prescribed spectral density. The system–bath contact is bilinear and *transverse* on the qubit,

$$H_{SB_\alpha} = \sigma_x^{(i(\alpha))} \otimes \sum_k \lambda_{\alpha k} (a_{\alpha k} + a_{\alpha k}^\dagger). \quad (5)$$

- *Why bilinear.* Leading order in a weak coupling expansion; higher powers are suppressed and give the Born approximation its control parameter.
- *Why σ_x and not σ_z .* To thermalise a qubit the bath must be able to *flip* it, $|0\rangle \leftrightarrow |1\rangle$, exchanging the energy quantum ϵ_i ; $\sigma_x = \sigma_+ + \sigma_-$ does exactly this. A σ_z coupling commutes with H_i and only *dephases*: it could never establish a thermal population and hence could not encode a temperature. (Contrast the *drive* of Sec. 3.2, which is chosen *longitudinal* σ_z precisely so that it does *not* pump populations.)
- *Detailed balance.* With spectral density $J_\alpha(\omega) = \pi \sum_k \lambda_{\alpha k}^2 \delta(\omega - \omega_{\alpha k})$, the golden-rule rates obey the KMS condition $\gamma_\alpha(-\omega) = e^{-\beta_\alpha \omega} \gamma_\alpha(\omega)$, which forces qubit $i(\alpha)$ toward the Gibbs occupation $g(\beta_\alpha \epsilon_i)$. These rates are precisely the γ, μ, μ' of the reset model.

The bath–to–qubit assignment, and the reason for each, is:

Bath	Qubit	Inverse temp.	Role
B_0	C_0	β_0 (fixed)	reference / perceptron bias
B_1	C_1	$\beta_1 \in \{\beta_{\text{hot}}, \beta_{\text{cold}}\}$	logical input channel
B_z	C_z	$\beta_z(t)$ (<i>finite</i> , cap. $\mathcal{C}$)	readout reservoir
B_r	$C_{\mathcal{M}}$	β_r (fixed)	modulator anchor

Why is B_z finite while B_0, B_1, B_r are infinite? An infinite reservoir has, by definition, a temperature that does not move—useless as an output, which must be a variable that *responds* to the computation. Endowing B_z with a finite heat capacity $\mathcal{C}$ promotes $\beta_z(t)$ to a dynamical calorimetric variable (Eq. (14)) that integrates the net heat current and settles at β_z^∞ : the finiteness of B_z is exactly what converts a heat *current* into a readable *temperature*.

(4) The modulator Hamiltonian: why one qubit, why two baths.

$$H_{\mathcal{M}} = \epsilon_{\mathcal{M}} n_{\mathcal{M}}, \quad \epsilon_{\mathcal{M}} = \epsilon_z, \quad (6)$$

in *simultaneous* contact with B_r (rate γ) and B_z (rate μ'):

$$H_{\mathcal{M}B_r} = \sigma_x^{(\mathcal{M})} \otimes \sum_k \lambda_{rk} (a_{rk} + a_{rk}^\dagger), \quad H_{\mathcal{M}B_z} = \sigma_x^{(\mathcal{M})} \otimes \sum_k \lambda_{zk} (a_{zk} + a_{zk}^\dagger). \quad (7)$$

- *Why resonant with C_z* ($\epsilon_{\mathcal{M}} = \epsilon_z$). The modulator must act on the *same* output channel that the collector drives, so it competes on equal spectral footing.
- *Why two baths.* A single-bath qubit merely thermalises to that bath and does nothing to the output. Straddling the fixed B_r and the output B_z , the modulator becomes a *heat valve* that continuously pulls β_z toward β_r (Eq. (12)) with strength μ' , competing with the collector current. That competition is what saturates the transfer curve into a sigmoid (Sec. 3.1.5); remove the second bath and the valve—and with it the whole nonlinearity—disappears, leaving the linear amplifier flagged in the overview.

(5) The Floquet extension $\Delta H_F(t)$: form and justification. For each inserted buffer $F \in \{\mathcal{F}_r, \mathcal{F}_z\}$ (Sec. 3.2), the extension both *adds*

$$\Delta H_F(t) = \underbrace{\frac{\epsilon_F(t)}{2} \sigma_z^{(F)}}_{H_F(t), \epsilon_F(t) = \epsilon_F + A \cos \Omega t} + \underbrace{g_F (\sigma_+^{(C)} \sigma_-^{(F)} + \text{h.c.})}_{H_{CF}} + \underbrace{\sigma_x^{(F)} \otimes \sum_k \lambda_k (a_k + a_k^\dagger)}_{H_{FB}}, \quad (8)$$

and *moves* the corresponding bath contact off the collector qubit onto the buffer (the collector reaches B_0/B_z only *through* F). Each piece is fixed by a design principle:

- *Longitudinal drive* ($\propto \sigma_z^{(F)}$, commuting with H_F): it does no work on buffer populations and only dresses the transition phases, giving the clean Bessel sideband comb (30). A transverse drive $\sigma_x \cos \Omega t$ would directly pump and heat the buffer and blur the comb.
- *Resonant exchange* H_{CF} (same RWA logic as H_{int}): transfers excitations between collector qubit and buffer at no bare-energy cost; counter-rotating $\sigma_x \sigma_x$ pieces are dropped.
- *Transverse buffer-bath* H_{FB} (same reason as (5)): the buffer must relax through its reservoir.
- *Only the reference and output branches are dressed.* Buffering the input branch C_1 – B_1 would alter the logical encoding (27); the input is left bare so the bit definition is untouched.

Summary: every term, and what is deliberately absent. Table 1 collects the ingredients. The final row lists the couplings a reader might expect but that are omitted by design, each for a reason given above.

Everything downstream is a reduction of this Hamiltonian: tracing out the reservoirs in the weak-coupling, secular limit yields the reset-model rates γ, μ, μ' and hence the currents (13)–(12); the population/virtual-temperature derivation of Sec. 3.1.2 is the steady state of the swap (3); and the buffer terms feed the sideband analysis of Sec. 3.2.1. A fully rigorous statement of the reduction, including the validity window and the induced Lamb shift, is Proposition 7.1 in Part II.

Term	Expression	Acts on / role and why this form
Free qubits H_S	$\sum_i \epsilon_i n_i$	C_0, C_1, C_z ; holds logical populations. Diagonal only—no transverse field, so the readout basis is an eigenbasis.
Collector H_{int}	$g(100\rangle\langle 011 + \text{h.c.})$	$C_0 C_1 C_z$; resonant SWAP of the virtual qubit into C_z . Three-body and secular (RWA); enforces $\epsilon_z = \epsilon_0 - \epsilon_1$.
Modulator $H_{\mathcal{M}}$	$\epsilon_z n_{\mathcal{M}}$	$C_{\mathcal{M}}$; heat-valve anchor, resonant with the output.
Reservoirs H_{B_α}	$\sum_k \omega_{\alpha k} a_{\alpha k}^\dagger a_{\alpha k}$	$B_{0,1,z,r}$; dissipation, input register, and power supply. Harmonic continuum.
System–bath H_{SB_α}	$\sigma_x^{(i)} \otimes \sum_k \lambda_{\alpha k} (a_{\alpha k} + a_{\alpha k}^\dagger)$	qubit + bath; thermalises populations. Transverse (σ_x) so the bath can flip the qubit; σ_z would only dephase.
Buffer drive $H_F(t)$	$\frac{\epsilon_F(t)}{2} \sigma_z^{(F)}$	F ; spectral filter. Longitudinal—no population work.
Buffer exchange H_{CF}	$g_F(\sigma_+^{(C)} \sigma_-^{(F)} + \text{h.c.})$	C, F ; mediates qubit–bath contact. Resonant (RWA).
Buffer–bath H_{FB}	$\sigma_x^{(F)} \otimes \sum_k \lambda_k (a_k + a_k^\dagger)$	F +bath; buffer relaxation. Transverse.
<i>Absent by design</i>	bare C_0 – C_1 exchange and dispersive $\sigma_z \sigma_z$ (would spoil $\tau_0 \otimes \tau_1$); counter-rotating three-body terms (non-resonant); transverse bare fields on machine qubits (would mix logical states); direct collector–bath contact on dressed branches; a buffer on the input branch (would change the encoding).	

Table 1: Complete term-by-term Hamiltonian of the (Floquet-buffered) thermodynamic neuron, Eq. (1). Each entry names the support, the physical role, and the design reason for its specific form; the last row records the couplings that are deliberately excluded.

3.1.2 The Collector architecture

Why need a collector? Consider a classical perceptron model given by :

$$y = \sum_i \omega_i x_i$$

before applying an activation function , the perceptron first computes the weighted sum of inputs. So the thermodynamic neuron is analogous to the perceptron model so instead of electrical voltage and the inputs are thermal reservoirs characterized by their inverse temperature.:

$$\beta_i = \frac{1}{K_\beta T_i}$$

Inside the collector contains 3 qubits C_0, C_1, C_z where C_0, C_1 are the input qubits with energy bandgap ϵ_1, ϵ_0 and the C_z is the output qubit with band gap ϵ_z .

Lets first do an numerical rigourous interpretation of the collector: So there are 2 input qubits C_1 and C_0 now thses qubits are connected 2 reservoirs B_0, B_1 amomgst whovht the bath connected to qubit 0 is a constant bath meaning the it acts a refrence , while thebath connected tot qubit 1 is through whoch tht e input is necoded into and sent to the output qubit. The thermal states of each baths are :

$$\tau_0 = \frac{e^{-\beta_0}}{Z_0}, \quad \tau_1 = \frac{e^{-\beta_1}}{Z_1}$$

For a single qubit the excitation population number is given by :

$$\mathcal{H}_i = \epsilon_i |1\rangle \langle 1|$$

where the excited propbabailty is

$$\mathcal{P}_i(1) = \frac{e^{-\beta_1 \epsilon_i}}{1 + \beta_i \epsilon_i}$$

and the ground state population is given by:

$$\mathcal{P}_i(0) = \frac{1}{1 + \beta_i \epsilon_i}$$

so the boltzman law states that:

$$\frac{\mathcal{P}(1)}{\mathcal{P}(0)} = e^{-\beta_i \epsilon_i}$$

Now for the combined system we do the same:

$$\tau_0 \otimes \tau_1$$

Now the basis states are

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

now the populations for each qubit is :

$$\mathcal{P}_{00} = p_0(0)p_1(0), \mathcal{P}_{01} = p_0(0)p_1(1), \mathcal{P}_{10} = p_0(1)p_1(0), \mathcal{P}_{11} = p_0(1)p_1(1),$$

In the above basis vector state of the system what it is done is that they have defined the states $|10\rangle, |01\rangle$ as the virtual qubits because unlike the other pure states they form a 2D subspace (the virtual qubit). So mathematically we define

$$|10\rangle = |1\rangle_v, |01\rangle = |0\rangle_v$$

Now calculating the virtual population of the virtual qubits

$$\mathcal{P}_v(1) = \mathcal{P}_{10}, \mathcal{P}_v(0) = \mathcal{P}_{01}$$

Now using the boltzmann ratio we get :

$$\frac{\mathcal{P}_v(0)}{\mathcal{P}_v(1)} = \frac{\mathcal{P}_0 \mathcal{P}_1}{\mathcal{P}_1 \mathcal{P}_0}$$

Using the Boltzmann relations

$$\frac{p_0(0)}{p_0(1)} = e^{-\beta_0 \epsilon_0},$$

and

$$\frac{p_1(0)}{p_1(1)} = e^{-\beta_1 \epsilon_1},$$

we obtain

$$\frac{p_1(1)}{p_1(0)} = e^{+\beta_1 \epsilon_1}.$$

Substituting into the population ratio yields

$$\frac{P_v(0)}{P_v(1)} = e^{-\beta_0 \epsilon_0} e^{+\beta_1 \epsilon_1}.$$

Hence,

$$\frac{P_v(0)}{P_v(1)} = e^{-(\beta_0 \epsilon_0 - \beta_1 \epsilon_1)}.$$

Now consider a genuine thermal qubit with energy gap E . Its thermal population ratio is

$$\frac{P(0)}{P(1)} = e^{-\beta E}.$$

The virtual subspace exhibits exactly the same functional form. We therefore define a virtual energy gap

$$E_v = \epsilon_0 - \epsilon_1,$$

and a virtual inverse temperature β_v through

$$\frac{P_v(0)}{P_v(1)} = e^{-\beta_v E_v}.$$

Comparing the exponents gives

$$\beta_v(\epsilon_0 - \epsilon_1) = \beta_0 \epsilon_0 - \beta_1 \epsilon_1.$$

Therefore,

$$\beta_v = \frac{\epsilon_0}{\epsilon_0 - \epsilon_1} \beta_0 - \frac{\epsilon_1}{\epsilon_0 - \epsilon_1} \beta_1$$

This quantity is called the virtual inverse temperature.

The terminology “virtual” is used because no physical thermal reservoir exists at temperature $T_v = 1/(k_B \beta_v)$. Rather, this effective temperature emerges from the population ratio

$$\frac{P_{01}}{P_{10}}.$$

The virtual qubit is therefore not a fundamental subsystem but an effective two-level system encoded within the composite Hilbert space of the two thermal qubits C_0 and C_1 .

To extract and utilize this virtual temperature, a third qubit C_z is introduced. Its energy gap is chosen to be resonant with the virtual energy gap:

$$\epsilon_z = \epsilon_0 - \epsilon_1 = E_v.$$

This C_z is connected to another bath B_z . so due to this creation of virtual temperature of the initial input qubits the output qubit temperature tries to equalize the virtual temperature or to be more precise the bath connected to the output qubit tries to equilibrate to this virtual temperature resulting in a linear response behaviour making it necessary for the presence of the Modulator.

3.1.3 Modulator

The modulator $\mathcal{M}$ consists of a single qubit $C_{\mathcal{M}}$ with energy gap chosen resonant with the output qubit,

$$\epsilon_{\mathcal{M}} = \epsilon_z,$$

placed in simultaneous contact with *two* baths: the reference bath B_r at fixed inverse temperature β_r (coupling rate γ) and the finite output bath B_z (coupling rate μ'). Within the reset model used for the collector, its state $\rho_{\mathcal{M}}$ obeys

$$\dot{\rho}_{\mathcal{M}} = \mathcal{L}^{(r)}[\rho_{\mathcal{M}}] + \mathcal{L}^{(z)}[\rho_{\mathcal{M}}] = \gamma(\tau(\beta_r) - \rho_{\mathcal{M}}) + \mu'(\tau(\beta_z) - \rho_{\mathcal{M}}). \quad (9)$$

Since both dissipators are diagonal in the energy basis, it suffices to track the excited population $p_{\mathcal{M}} := \langle 1 | \rho_{\mathcal{M}} | 1 \rangle$:

$$\dot{p}_{\mathcal{M}} = \gamma [g(\beta_r \epsilon_z) - p_{\mathcal{M}}] + \mu' [g(\beta_z \epsilon_z) - p_{\mathcal{M}}]. \quad (10)$$

Setting $\dot{p}_{\mathcal{M}} = 0$ gives the exact steady-state population

$$p_{\mathcal{M}}^{ss} = \frac{\gamma g_z(\beta_r) + \mu' g_z(\beta_z)}{\gamma + \mu'} \xrightarrow{\mu' \ll \gamma} g_z(\beta_r), \quad (11)$$

i.e. in the regime $\mu' \ll \gamma$ the modulator qubit is pinned to the reference temperature β_r , essentially unperturbed by the output bath. The heat current it delivers to B_z under the reset model is then

$$j_{\mathcal{M}} = \mu' \epsilon_z [g_z(\beta_z) - g_z(\beta_r)]. \quad (12)$$

The modulator therefore continuously pulls β_z toward β_r with strength μ' . This is the entire mechanism: it is a *single-qubit heat valve* between B_r and B_z that competes with the collector current. The two free parameters (β_r, μ') will be fixed below so that the competition confines β_z^∞ to a designated logical window, which is precisely what converts the linear amplifier of Sec. 3.1.2 into a saturating, sigmoid-like response.

3.1.4 Dynamics of the full machine: two time scales

Collecting the two currents into B_z — the collector current

$$j_{\mathcal{C}} = \mu \epsilon_z [g_z(\beta_z) - g_z(\beta_v)], \quad (13)$$

derived in Sec. 3.1.2 from the virtual-qubit resonance, and the modulator current (12) — the finite bath B_z with heat capacity $\mathcal{C}$ evolves according to the calorimetric equation $\dot{T}_z = (j_{\mathcal{C}} + j_{\mathcal{M}})/\mathcal{C}$, which in terms of $\beta_z = 1/(k_B T_z)$ reads

$$\dot{\beta}_z = -\frac{k_B \beta_z^2}{\mathcal{C}} (j_{\mathcal{C}} + j_{\mathcal{M}}). \quad (14)$$

The consistency of using the *steady-state* expressions (13) and (12) inside (14) rests on a separation of time scales enforced by construction,

$$\gamma \gg \mu, \mu', \quad (15)$$

so that the internal (fast) relaxation of $\mathcal{C}$ and $\mathcal{M}$ is slaved to the instantaneous value of $\beta_z(t)$: on the fast scale C_z thermalizes to β_v and $C_{\mathcal{M}}$ to β_r , while on the slow scale $\beta_z(t)$ drifts under the residual currents. This is an adiabatic elimination of the machine's internal degrees of freedom and is the autonomous analogue of the quasi-static gate assumption in CMOS analysis.

3.1.5 Steady state and the activation function

The nonequilibrium steady state of the output is defined by $\dot{\beta}_z = 0$, i.e. $j_{\mathcal{C}} + j_{\mathcal{M}} = 0$. Substituting (13) and (12),

$$(\mu + \mu') g_z(\beta_z^\infty) = \mu g_z(\beta_v) + \mu' g_z(\beta_r), \quad (16)$$

hence, defining the branching ratio $\Delta := \mu/(\mu + \mu')$,

$$\boxed{g_z(\beta_z^\infty) = \Delta g_z(\beta_v) + (1 - \Delta) g_z(\beta_r)} \quad (17)$$

The steady-state excitation of the output channel is a convex combination of the collector's target (β_v) and the modulator's anchor (β_r), weighted purely by coupling rates. Inverting $g_z(\beta) = (1 + e^{\beta \epsilon_z})^{-1}$,

$$\beta_z^\infty = \frac{1}{\epsilon_z} \ln \left[\frac{1}{\Delta g_z(\beta_v) + (1 - \Delta) g_z(\beta_r)} - 1 \right]. \quad (18)$$

Fixing the modulator: confinement to the logical window. For β_z^∞ to serve as a logical signal we demand that it be confined to a window $[\beta_{\text{hot}}, \beta_{\text{cold}}]$ (with $\beta_{\text{cold}} > \beta_{\text{hot}}$) even when the virtual temperature saturates. Since g_z is monotonically decreasing, $\beta_v \rightarrow +\infty$ gives $g_z(\beta_v) \rightarrow 0$ and $\beta_v \rightarrow -\infty$ gives $g_z(\beta_v) \rightarrow 1$. Imposing

$$\beta_v \rightarrow +\infty : \quad (1 - \Delta) g_z(\beta_r) \stackrel{!}{=} g_z(\beta_{\text{cold}}), \quad (19)$$

$$\beta_v \rightarrow -\infty : \quad \Delta + (1 - \Delta) g_z(\beta_r) \stackrel{!}{=} g_z(\beta_{\text{hot}}), \quad (20)$$

and subtracting (19) from (20),

$$\Delta = g_z(\beta_{\text{hot}}) - g_z(\beta_{\text{cold}}), \quad g_z(\beta_r) = \frac{g_z(\beta_{\text{cold}})}{1 - \Delta}. \quad (21)$$

These two design equations fix $(\mu'/\mu, \beta_r)$ in terms of the chosen logical window. Substituting back into (18),

$$\beta_z^\infty = \frac{1}{\epsilon_z} \ln[Q(\beta_v)^{-1} - 1], \quad Q(\beta_v) := g_z(\beta_{\text{hot}}) g_z(\beta_v) + g_z(\beta_{\text{cold}})(1 - g_z(\beta_v)). \quad (22)$$

Equation (22) is the exact transfer characteristic of the neuron under the reset model.

Emergence of the sigmoid. The connection to the perceptron activation is transparent once one notices an *exact* identity. From the definition of the virtual temperature, $\beta_v \epsilon_z = \beta_0 \epsilon_0 - \beta_1 \epsilon_1$, so with $f(x) := (1 + e^x)^{-1}$,

$$g_z(\beta_v) = \frac{1}{1 + e^{\beta_v \epsilon_z}} = f(\beta_0 \epsilon_0 - \beta_1 \epsilon_1) \quad (\text{no approximation}). \quad (23)$$

The sigmoid is thus already present at the level of the virtual-qubit population; the modulator's role is to map it affinely onto the temperature axis. Indeed, expanding (22) for small ϵ_z (with $\beta_{\text{hot}}, \beta_{\text{cold}}$ of order unity): using $g_z(\beta) = \frac{1}{2} - \frac{\beta \epsilon_z}{4} + \mathcal{O}(\epsilon_z^3)$,

$$Q(\beta_v) = \frac{1}{2} - \frac{\epsilon_z}{4} [\beta_{\text{cold}} - (\beta_{\text{cold}} - \beta_{\text{hot}}) g_z(\beta_v)] + \mathcal{O}(\epsilon_z^2), \quad (24)$$

$$\begin{aligned} \beta_z^\infty &= \frac{1}{\epsilon_z} \ln \frac{1 - Q}{Q} = \frac{1}{\epsilon_z} \ln \left[1 + \epsilon_z (\beta_{\text{cold}} - (\beta_{\text{cold}} - \beta_{\text{hot}}) g_z(\beta_v)) + \mathcal{O}(\epsilon_z^2) \right] \\ &= \beta_{\text{hot}} g_z(\beta_v) + \beta_{\text{cold}} (1 - g_z(\beta_v)) + \mathcal{O}(\epsilon_z). \end{aligned} \quad (25)$$

Equations (23)–(25) together give

$$\beta_z^\infty \simeq \beta_{\text{cold}} + (\beta_{\text{hot}} - \beta_{\text{cold}}) f(\epsilon_0(\beta_0 - \beta_1) + \beta_1 \epsilon_z), \quad (26)$$

a sigmoidal interpolation between the two logical rails whose argument is the perceptron pre-activation: weight ϵ_1 (steepness $\propto \epsilon_0 \approx \epsilon_1$ for small ϵ_z), bias set by β_0 . In the limit $\epsilon_1 \rightarrow \infty$ the transfer characteristic approaches the ideal inverted step function; the price, as quantified by the entropy production analysis of [2], is monotonically increasing dissipation — the error–dissipation tradeoff that Part II, Conjecture 9.1, generalizes to the driven setting.

3.1.6 Logic encoding and the NOT operation

Logical values are encoded and decoded via

$$x = \begin{cases} 0, & \beta_1 = \beta_{\text{hot}}, \\ 1, & \beta_1 = \beta_{\text{cold}}, \end{cases} \quad y = \begin{cases} 0, & \beta_z^\infty \leq (1 + \delta) \beta_{\text{hot}}, \\ 1, & \beta_z^\infty \geq (1 - \delta) \beta_{\text{cold}}, \\ \emptyset & \text{otherwise,} \end{cases} \quad (27)$$

where δ is a tolerance defining guard bands around the rails and $\emptyset$ marks an invalid readout. The inversion follows directly from the sign structure of β_v : for a hot input ($\beta_1 = \beta_{\text{hot}} < \beta_0$) the collector operates as a refrigerator, β_v is large, $g_z(\beta_v) \rightarrow 0$, and (25) gives $\beta_z^\infty \rightarrow \beta_{\text{cold}}$ ($y = 1$); for a cold input the collector is a heat pump, $g_z(\beta_v) \rightarrow 1$, and $\beta_z^\infty \rightarrow \beta_{\text{hot}}$ ($y = 0$). Fluctuations of the finite bath around β_z^∞ (Gaussian with width controlled by $\mathcal{C}$) produce a nonzero error probability $\langle \xi \rangle$; it is exactly this readout fragility — logical information carried by a fluctuating classical temperature at the end of an open, bidirectional channel — that motivates the buffer architecture of the next subsection.

3.2 Insertion of the Floquet buffers

We now make the proposal of Sec. 1.2 concrete. Two buffers are inserted:

- $\mathcal{F}_r$ in the *reference branch*, mediating $C_0 \leftrightarrow \mathcal{F}_r \leftrightarrow B_0$;
- $\mathcal{F}_z$ in the *output branch*, mediating $C_z \leftrightarrow \mathcal{F}_z \leftrightarrow B_z$.

The input branch $C_1 - B_1$ is left undressed so that the logical encoding is untouched. Each buffer is a two-level system with a periodically modulated gap,

$$H_{\mathcal{F}}(t) = \frac{\epsilon_F(t)}{2} \sigma_z^{(F)}, \quad \epsilon_F(t) = \epsilon_F + A \cos(\Omega t), \quad H_{\mathcal{F}}(t+T) = H_{\mathcal{F}}(t), \quad T = \frac{2\pi}{\Omega}, \quad (28)$$

coupled (i) to its neighbouring collector qubit by a resonant excitation-exchange term of strength g_F , $H_{CF} = g_F(\sigma_+^{(C)} \sigma_-^{(F)} + \text{h.c.})$, and (ii) to its bath by a standard bilinear coupling $H_{FB} = \sigma_x^{(F)} \otimes \sum_k \lambda_k (a_k + a_k^\dagger)$ with spectral density $J(\omega)$ and KMS-obeying rates $\gamma(\omega)$, i.e. $\gamma(-\omega) = e^{-\beta\omega} \gamma(\omega)$. The modulated-gap (longitudinal) drive is chosen deliberately: it commutes with $\sigma_z^{(F)}$, so the drive does no work on the buffer populations directly; it only dresses the *phases* of the transition operators, which is exactly the handle needed for spectral filtering.

3.2.1 Sideband decomposition of the buffer–bath coupling

In the interaction picture with respect to $H_{\mathcal{F}}(t)$, the lowering operator of the buffer accumulates the integrated phase

$$\sigma_-^{(F)}(t) = \sigma_-^{(F)} \exp\left[-i \int_0^t \epsilon_F(t') dt'\right] = \sigma_-^{(F)} e^{-i\epsilon_F t} e^{-i\frac{A}{\Omega} \sin(\Omega t)}. \quad (29)$$

The Jacobi–Anger expansion $e^{-iz \sin \theta} = \sum_{q \in \mathbb{Z}} J_q(z) e^{-iq\theta}$ (with J_q the Bessel function of the first kind) resolves this into a sideband comb:

$$\sigma_-^{(F)}(t) = \sigma_-^{(F)} \sum_{q \in \mathbb{Z}} J_q\left(\frac{A}{\Omega}\right) e^{-i(\epsilon_F + q\Omega)t} \quad (30)$$

The single transition line at ϵ_F is replaced by lines at $\epsilon_F + q\Omega$ with weights $J_q^2(A/\Omega)$ — this is the quantum-optical content of the FM analogy of Sec. 2.1: the modulation index β_{FM} of the communication example is literally A/Ω here, and the Carson bandwidth of FM corresponds to the width of the populated sideband comb. Inserting (30) into the second-order (Floquet–Born–Markov) golden-rule rates and applying the secular approximation with respect to the drive (valid for $\Omega \gg \gamma(\epsilon_F)$, so cross-sideband interference terms average out), the downward and upward rates on the buffer become

$$\Gamma_\downarrow = \sum_q J_q^2\left(\frac{A}{\Omega}\right) \gamma(\epsilon_F + q\Omega), \quad (31)$$

$$\Gamma_\uparrow = \sum_q J_q^2\left(\frac{A}{\Omega}\right) \gamma(-\epsilon_F - q\Omega) = \sum_q J_q^2\left(\frac{A}{\Omega}\right) e^{-\beta(\epsilon_F + q\Omega)} \gamma(\epsilon_F + q\Omega), \quad (32)$$

where the last equality used the KMS condition of the physical bath at inverse temperature β . Equations (31)–(32) are the Tien–Gordon structure of photon-assisted transport [10, 11]: the driven buffer samples the bath spectral function at a comb of frequencies, with Bessel weights controlled by the single dimensionless knob A/Ω .

Two filtering mechanisms follow immediately. (i) *Line suppression*: at the zeros of J_0 ($A/\Omega \approx 2.405, 5.520, \dots$) the carrier transition at ϵ_F is dynamically switched off — coherent destruction of the direct thermal channel — while sidebands remain open. (ii) *Spectral relocation*: by choosing Ω so that $\epsilon_F + q^*\Omega$ sits where $J(\omega)$ is large (or in a gap), a single sideband q^* can be made to dominate the sums.

3.2.2 Floquet-synthesized effective temperature

The driven buffer does not merely rescale the coupling; it changes the *detailed-balance ratio* seen at the buffer gap. Define the effective inverse temperature β_{eff} of the dressed environment by

$$e^{-\beta_{\text{eff}} \epsilon_F} := \frac{\Gamma_{\uparrow}}{\Gamma_{\downarrow}} = \frac{\sum_q J_q^2(A/\Omega) e^{-\beta(\epsilon_F + q\Omega)} \gamma(\epsilon_F + q\Omega)}{\sum_q J_q^2(A/\Omega) \gamma(\epsilon_F + q\Omega)}. \quad (33)$$

For $A = 0$ only $q = 0$ survives and $\beta_{\text{eff}} = \beta$, as it must. In the single-sideband regime where q^* dominates,

$$\beta_{\text{eff}} \approx \beta \frac{\epsilon_F + q^*\Omega}{\epsilon_F}, \quad (34)$$

so the same physical bath at β is presented to the collector as an effective bath that is colder ($q^* > 0$), hotter ($-\epsilon_F < q^*\Omega < 0$), or even population-inverted ($q^*\Omega < -\epsilon_F$, $\beta_{\text{eff}} < 0$). In general (33) is not of Gibbs form across different gaps — the dressed environment is athermal — but on the single relevant transition of C_0 (which exchanges excitations resonantly with $\mathcal{F}_r$ at rate g_F , and is therefore slaved to the buffer’s steady population in the regime $g_F \gg \Gamma_{\downarrow} + \Gamma_{\uparrow} \gg \mu$) it acts exactly as a thermal reservoir at β_{eff} . A rigorous derivation of this statement, including the conditions under which the composite $\mathcal{F}+B$ admits Markovian elimination and the form of the induced Lamb shift, is the content of Proposition 7.1 in Part II; here we take the physical picture and draw its consequences for the neuron.

3.2.3 Consequence I: drive-tunable weights and bias

Replacing $\beta_0 \rightarrow \beta_{\text{eff}}(A/\Omega, \Omega)$ in the virtual temperature,

$$\beta_v^{\text{eff}} = \frac{\epsilon_0}{\epsilon_0 - \epsilon_1} \beta_{\text{eff}}\left(\frac{A}{\Omega}, \Omega\right) - \frac{\epsilon_1}{\epsilon_0 - \epsilon_1} \beta_1, \quad (35)$$

the reference-branch buffer promotes the perceptron *bias* from a hardware constant to a runtime control parameter. In the autonomous neuron the weights and bias are frozen into the fabricated energy gaps ϵ_i and the physical reference temperature β_0 ; training or reconfiguring the gate means re-engineering the hardware. With $\mathcal{F}_r$ in place, the threshold location of the activation (25) — which sits at $\beta_1 \approx \beta_{\text{eff}}$ for small ϵ_z — is swept electrically/optically through A/Ω without touching any qubit gap. The same neuron can thus be repurposed (e.g. shifting the NOT threshold, or, in the multi-input generalization, retraining a linearly separable function *in situ*) at the cost of drive work. This is, to our knowledge, the feature that most sharply distinguishes the buffered architecture from the autonomous one: it trades autonomy for programmability, in direct analogy with the gate terminal of a transistor.

3.2.4 Consequence II: output-branch buffer as a gated, rectifying readout stage

The output buffer $\mathcal{F}_z$ addresses the readout fragility identified in Sec. 3.1.6. Two mechanisms operate.

(a) Spectral rectification (suppressing backflow). The current from B_z back into the collector is carried by the same comb (31)–(32). Because absorption and emission sample $\gamma(\pm(\epsilon_F + q\Omega))$ asymmetrically whenever the comb straddles regions where $J(\omega)$ varies strongly, the ratio of forward (collector $\rightarrow B_z$) to backward conductance can be engineered away from unity. Operating $\mathcal{F}_z$ near a J_0 zero with a single dominant blue sideband placed at a peak of $J(\omega)$ makes the effective channel predominantly emissive into B_z : the buffer behaves as an energy diode for the readout current. Within the effective-rate description, this replaces the symmetric reset coupling μ of Eq. (13) by an asymmetric pair $(\mu_{\rightarrow}, \mu_{\leftarrow})$ with $\mu_{\leftarrow}/\mu_{\rightarrow} < 1$, which steepens the approach of $\beta_z(t)$ to its fixed point and reduces the sensitivity of β_z^∞ to fluctuations of the finite bath — i.e. it widens the usable guard band δ in (27) at fixed error rate.

(b) Stroboscopic gating. Since the comb weights depend on A/Ω , ramping A between A_{on} and a J_0 zero periodically opens and closes the $C_z - B_z$ channel in synchrony with an external clock. The neuron then integrates its computation while the channel is shut (no information leaks to, and no noise enters from, B_z) and commits the result during short readout windows. This is precisely the clock-gateway mechanism that Ref. [2] introduces externally to concatenate neurons into networks (their Supplementary Material C), here realized *inside* the machine by the drive itself: t_{steady} and t_{const} of their protocol map onto the closed and open fractions of the drive period. The buffered neuron is therefore natively composable, at the price of no longer being autonomous.

3.2.5 Cost accounting and outlook of Part 1

Both consequences are paid for in work. The drive injects average power $P_{\text{drive}}(t) = \langle \partial_t H_{\mathcal{F}}(t) \rangle$, and every sideband-assisted exchange of a bath quantum $\epsilon_F + q\Omega$ transfers $q\Omega$ from the drive per event; the effective-temperature shift (34) is thus a work-powered refrigeration or pumping effect, consistent with the second law for periodically driven open systems [5]. Whether the gain in logical distinguishability can outrun this cost — and what the optimal exchange rate is — is exactly the work-distinguishability tradeoff formalized as Conjecture 9.1 and probed numerically in the staged programme of Part II, Sec. 10. The weak-coupling regime in which the effective-rate picture of this section is quantitatively valid is delimited by Proposition 7.1; the non-Markovian regime, where the buffer correlation time competes with $1/g_F$ and the sideband picture breaks down, requires the chain-mapped MPS machinery of Sec. 8.

3.3 Validity window, breakdown, and numerical stability

3.3.1 Hierarchy of scales and the validity window

The effective-rate picture of Secs. 3.2.1–3.2.2 is a chain of three eliminations, each with its own small parameter. Denote by

$$\Gamma_F := \Gamma_{\downarrow} + \Gamma_{\uparrow} = \sum_q J_q^2 \left(\frac{A}{\Omega} \right) \left[1 + e^{-\beta(\epsilon_F + q\Omega)} \right] \gamma(\epsilon_F + q\Omega) \quad (36)$$

the total dressed linewidth of the buffer, i.e. the inverse of its correlation time $\tau_c = 1/\Gamma_F$ in the periodic steady state. The three approximations and their control parameters are:

(V1) **Drive-secular approximation.** Cross-sideband interference terms in the dissipator oscillate at multiples of Ω and average out over the buffer relaxation time provided

$$\eta_1 := \frac{\Gamma_F}{\Omega} \ll 1. \quad (37)$$

When $\eta_1 \sim 1$ the sidebands overlap within their linewidth: the comb (30) merges into a continuum and the weights J_q^2 lose meaning individually.

(V2) **Born–Markov elimination of the buffer.** Treating the buffer+bath compound as a Markovian environment for C_0 requires its correlation time to be short compared to the induced relaxation of C_0 . The induced rate is of order g_F^2/Γ_F (second-order golden rule against a Lorentzian of width Γ_F), so Markovianity requires

$$\eta_2 := \frac{g_F}{\Gamma_F} \ll 1, \quad \text{relative error of the Lindblad reduction} = \mathcal{O}(\eta_2^2). \quad (38)$$

The error estimate follows from the Nakajima–Zwanzig expansion: the first neglected (fourth-order) contribution to the memory kernel is suppressed relative to the second-order one by $(g_F\tau_c)^2 = \eta_2^2$.

(V3) **Slaving of the slow calorimetric variable.** The dressed channel into B_z must remain the slowest process,

$$\eta_3 := \frac{\mu_{\text{eff}}}{g_F^2/\Gamma_F} \ll 1, \quad \mu_{\text{eff}} := \text{effective } C_z\text{--}B_z \text{ rate through } \mathcal{F}_z. \quad (39)$$

The window in which every result of Secs. 3.2.1–3.2.4 is quantitatively controlled is therefore the nested hierarchy

$$\boxed{\mu_{\text{eff}} \ll \frac{g_F^2}{\Gamma_F} \ll \Gamma_F \ll \Omega \lesssim \epsilon_F} \quad (40)$$

with cumulative error $\mathcal{O}(\eta_1) + \mathcal{O}(\eta_2^2) + \mathcal{O}(\eta_3)$. Note the structural tension already flagged in Proposition 7.1: making the buffer–bath coupling weaker (to protect the buffer’s periodic steady state) *lengthens* τ_c and therefore shrinks the admissible g_F through (38). The window (40) is real but narrow; everything outside it belongs to the non-Markovian programme.

3.3.2 The buffer as a tunable filtered environment

The physical content of the elimination (V2) is an effective spectral density. Linearizing the buffer around low excitation ($\sigma_-^{(F)} \rightarrow b$, valid for $\Gamma_\uparrow/\Gamma_F \ll 1$, i.e. an effectively cold dressed buffer; for larger thermal occupancy the qubit nonlinearity adds a saturation correction of order $\Gamma_\uparrow/\Gamma_F$), the buffer is a damped mode and the compound $\mathcal{F}+B$ presents to C_0 the Lorentzian-comb spectral density

$$J_{\text{eff}}(\omega) = \frac{g_F^2}{\pi} \sum_{q \in \mathbb{Z}} J_q^2\left(\frac{A}{\Omega}\right) \frac{\Gamma_F/2}{(\omega - \epsilon_F - q\Omega)^2 + (\Gamma_F/2)^2}. \quad (41)$$

Equation (41) makes the “spectral and temporal filter” statement of Sec. 1.2 quantitative: the drive relocates the weight of the comb (q , controlled by Ω), reweights it (J_q^2 , controlled by A/Ω), and the buffer–bath coupling sets the tooth width Γ_F . Markovianity of the reduced C_0 dynamics is the statement that J_{eff} is flat across the linewidth g_F^2/Γ_F of the qubit — automatically satisfied inside (40) because linewidth/toothwidth $= \eta_2^2$.

The corresponding memory kernel of the Nakajima–Zwanzig equation for C_0 ,

$$K(\tau) \propto g_F^2 e^{-\Gamma_F\tau/2} \sum_q J_q^2\left(\frac{A}{\Omega}\right) \cos[(\epsilon_F + q\Omega)\tau], \quad \tau \geq 0, \quad (42)$$

decays exponentially with rate $\Gamma_F/2$ — Assumption (A2) of Part II holds with $\tau_c = 2/\Gamma_F$. **Breakdown** occurs when $g_F \gtrsim \Gamma_F$: the poles of the C_0 –buffer propagator split by $\pm g_F$ (vacuum-Rabi doublet), $K(\tau)$ acquires undamped oscillations at g_F , the time-convolutionless expansion ceases to converge, and information flows back from the buffer into C_0 within a drive

period. A convenient numerical diagnostic is the BLP trace-distance witness: propagate two initial states $\rho_{1,2}(0)$ of C_0 and monitor

$$\sigma(t) := \frac{d}{dt} D(\rho_1(t), \rho_2(t)); \quad \sigma(t) > 0 \Rightarrow \text{information backflow} \Rightarrow \text{sideband picture invalid at } t. \quad (43)$$

$\sigma(t) > 0$ episodes are exactly the entropy-production backflow pathology described in Sec. 1.1, now with a measurable marker; the buffered design succeeds precisely to the extent that the drive parameters can push $\max_t \sigma(t) \rightarrow 0^+$ on the collector while retaining gain.

3.3.3 Numerical stability of the multiscale problem

The buffered neuron is numerically stiff by construction: the rate spectrum spans $\mu_{\text{eff}} \sim 10^{-4} \ll \gamma, \chi \sim 1 \ll \Omega \sim 10\text{--}10^2$ (in units of ϵ_z^{-1} taken as 1), a stiffness ratio $\kappa = \Omega/\mu_{\text{eff}} \gtrsim 10^5$. A naive explicit integration of the full master equation must resolve $h \lesssim 2\pi/(10\Omega)$ over a horizon $t_{\text{slow}} \sim \mathcal{C}/\mu_{\text{eff}}$, i.e. $\gtrsim 10^7\text{--}10^9$ steps with error accumulation that destroys the small currents $j_C + j_M \rightarrow 0$ near the fixed point. The stable scheme mirrors the physics (Sec. 3.1.4): exploit the same time-scale separation the machine itself uses.

Two-tier integrator.

1. **Fast tier (per grid point).** Freeze β_z . Solve the Floquet–Lindblad equation for the machine qubits over one period T to its periodic steady state (QuTiP: `FloquetBasis` / `fmmsolve`; or evolve $e^{\mathcal{L}_{\text{eff}}T}$ and take its Perron eigenvector). Extract the period-averaged currents $\bar{j}_C(\beta_z)$, $\bar{j}_M(\beta_z)$, $\bar{P}_{\text{drive}}(\beta_z)$.
2. **Slow tier.** Integrate the scalar ODE

$$\dot{\beta}_z = -\frac{k_B \beta_z^2}{\mathcal{C}} [\bar{j}_C(\beta_z) + \bar{j}_M(\beta_z)] \quad (44)$$

with any $\mathcal{O}(h^4)$ nonstiff method, interpolating $\bar{j}(\beta_z)$ from a precomputed grid ($\sim 30\text{--}50$ points suffice; the currents are smooth, monotone functions of $g_z(\beta_z)$ by (13),(12)).

The scheme replaces 10^8 coupled steps by $\mathcal{O}(10^2)$ fast-tier solves; its systematic error is the adiabatic-elimination error $\mathcal{O}(\mu_{\text{eff}}/\Gamma_F)$ — the same order as the modelling error already accepted in Sec. 3.1.4, so nothing is lost.

Structure preservation. Three invariants should be monitored, not assumed: (i) *complete positivity* — use the secular Floquet–Lindblad generator (CP by construction) and, when time-stepping $\mathcal{L}_{\text{eff}}(t)$ directly, a commutator-free Magnus (CFET) or Trotter splitting with $h\Omega \leq 2\pi/10$, checking $\lambda_{\min}(\rho) \geq -10^{-10}$; (ii) *trace* — drift $|\text{Tr } \rho - 1|$ bounds integrator error; (iii) *first law per cycle* — $\bar{W} + \sum_{\alpha} \bar{J}_{\alpha} = \Delta E_{\text{sys}}/T$ must close to integrator tolerance; violation localizes coding errors in the current operators before any physics is trusted.

Chain-MPS tier (beyond the window (40)). For $g_F \gtrsim \Gamma_F$ the fast tier is replaced by the chain-mapped MPS evolution of Sec. 8. Two additional stability constraints are specific to that representation:

$$t_{\text{sim}} < t_{\text{rec}} \approx \frac{N}{v_{\text{LR}}}, \quad v_{\text{LR}} \approx 2 \max_k t_k, \quad (45)$$

i.e. the chain length N must outrun the Lieb–Robinson cone so that no reflected excitation returns to the buffer within the simulated horizon (a finite chain is a perfect bath only until its own recurrence); and the time step must resolve the drive, $h \leq 2\pi/(10\Omega)$, or the Bessel weights alias — a driven Hamiltonian sampled below the Nyquist rate of Ω produces spurious sidebands indistinguishable from physics. Convergence is certified by joint refinement in $(\chi, d_{\text{Fock}}, N, h)$ with the period-averaged currents and P_{dist} as the target observables.

3.4 Exact correspondence with the CMOS inverter

The CMOS analogy of Sec. 1.2 can be promoted from metaphor to a quantitative dictionary, because both devices are fully characterized by a static transfer characteristic plus an isolation/clocking mechanism. Throughout, $\Delta\beta := \beta_{\text{cold}} - \beta_{\text{hot}}$ plays the role of the supply swing V_{DD} .

3.4.1 Voltage transfer characteristic and small-signal gain

The inverter’s figure of merit is the slope of its transfer curve at the switching threshold. Differentiating the activation (25) with $g_z(\beta_v) = f(\beta_0\epsilon_0 - \beta_1\epsilon_1)$ and $f'(x) = -f(x)(1 - f(x))$:

$$\mathcal{A}(\beta_1) := \frac{d\beta_z^\infty}{d\beta_1} = (\beta_{\text{hot}} - \beta_{\text{cold}}) f'(x) \frac{dx}{d\beta_1} = -\Delta\beta \epsilon_1 f(x)(1 - f(x)), \quad x = \beta_0\epsilon_0 - \beta_1\epsilon_1. \quad (46)$$

At the switching threshold $x = 0$ (input $\beta_1^* = \beta_0\epsilon_0/\epsilon_1 = \beta_0(1 + \epsilon_z/\epsilon_1) \approx \beta_0$), $f(1 - f) = 1/4$ and

$$\mathcal{A}_{\text{max}} = -\frac{\epsilon_1 \Delta\beta}{4} \xrightarrow{\Delta\beta=1} -\frac{\epsilon_1}{4}, \quad (47)$$

reproducing (for small ϵ_z , where $\epsilon_0 \approx \epsilon_1$) the slope computed in the Supplementary Material of [2]. The qubit gap ϵ_1 is the transconductance of the device: gain grows linearly in the energy invested per logical event, which is the microscopic origin of the gain–dissipation tradeoff (Fig. 3C of [2]) and of Conjecture 9.1.

3.4.2 Noise margins

CMOS noise margins are defined from the unity-gain points of the VTC. Imposing $|\mathcal{A}(\beta_1)| = 1$ in (46) gives $f(1 - f) = 1/(\epsilon_1 \Delta\beta)$, whose solutions for $\epsilon_1 \Delta\beta \gg 4$ are $f \approx 1/(\epsilon_1 \Delta\beta)$ and $f \approx 1 - 1/(\epsilon_1 \Delta\beta)$, i.e.

$$x_\pm = \pm \ln(\epsilon_1 \Delta\beta - 1) \approx \pm \ln(\epsilon_1 \Delta\beta). \quad (48)$$

The width of the transition region on the input axis and the resulting noise margins are therefore

$$W_{\text{trans}} = \frac{2 \ln(\epsilon_1 \Delta\beta)}{\epsilon_1}, \quad \text{NM}_H \approx \text{NM}_L \approx \frac{\Delta\beta}{2} - \frac{\ln(\epsilon_1 \Delta\beta)}{\epsilon_1}. \quad (49)$$

As $\epsilon_1 \rightarrow \infty$, $W_{\text{trans}} \rightarrow 0$ and $\text{NM} \rightarrow \Delta\beta/2$ — the ideal regenerative inverter — at logarithmically slow rate: doubling the noise margin deficit costs an e -fold in ϵ_1 , hence (by the entropy-production scaling) an e -fold in dissipation. Equation (49) is the thermodynamic-neuron analogue of the CMOS result that noise margins approach $V_{DD}/2$ as gain diverges, and it fixes the tolerance δ of the decoder (27) non-arbitrarily: $\delta \beta_{\text{hot}}, \delta \beta_{\text{cold}} < \text{NM}$.

3.4.3 The correspondence table

The one structural *disanalogy* deserves emphasis: a static CMOS inverter draws (ideally) zero supply current in either logic state, because one transistor of the complementary pair is cut off. The autonomous neuron has no cutoff — it is a NESS device and dissipates continuously (the “static leakage” row is not small). The gated Floquet buffer partially restores the CMOS advantage: with the $C_z - B_z$ channel closed at a J_0 zero between readout windows, the steady heat throughput of the output branch is suppressed, at the price of the clock work $\bar{W}$ — the same static-vs-dynamic power tradeoff that CMOS design negotiates with clock gating.

3.4.4 Architecture and correspondence diagrams

	CMOS inverter	(Buffered) thermodynamic neuron
Rails	V_{DD} , GND	$\beta_{\text{hot}}, \beta_{\text{cold}}$ (infinite baths)
Input node	gate voltage V_{in}	input inverse temperature β_1
Output node	drain node V_{out}	finite-bath temperature β_z^∞
Pull-down (nMOS on, V_{in} high)	output $\rightarrow$ GND	refrigerator regime (β_1 hot, β_v large): output $\rightarrow \beta_{\text{cold}}$
Pull-up (pMOS on, V_{in} low)	output $\rightarrow V_{DD}$	heat-pump regime (β_1 cold): output $\rightarrow \beta_{\text{hot}}$
Rail-to-rail restoration	complementary pair in cut-off/ohmic	modulator saturation, Eqs. (19)–(21)
Switching threshold V_M	device ratio W_p/W_n	reference temperature: $\beta_1^* \approx \beta_0$
Small-signal gain A_v at V_M	$g_m(r_{o,n} r_{o,p})$	$\mathcal{A}_{\text{max}} = -\epsilon_1 \Delta\beta/4$, Eq. (47)
Transconductance	g_m	qubit gap ϵ_1
Noise margins	from unity-gain points of VTC	Eq. (49)
Input–output isolation	MOS gate oxide (no DC gate current)	Floquet buffer $\mathcal{F}_z$: sideband rectification ($\mu_{\leftarrow} < \mu_{\rightarrow}$)
Dynamic (clocked) logic	precharge / evaluate phases, clock power $\propto CV_{DD}^2 f$	J_0 -zero gating of $\mathcal{F}_z$; drive work $\overline{W}$ per cycle
Body/threshold tuning (V_{th} shift)	back-gate bias	$\beta_0 \rightarrow \beta_{\text{eff}}(A/\Omega)$, Eq. (35)
Static leakage	subthreshold conduction	residual currents at $j_C + j_M = 0$: entropy production $\dot{\Sigma} > 0$ in the NESS

Table 2: Dictionary between the static CMOS inverter and the (Floquet-buffered) thermodynamic neuron. Every row is an identification of measured quantities, not an analogy of convenience: both columns are determined by the respective transfer characteristics and isolation mechanisms.

Part II

Abhishek: Periodically Driven Open-System Logic Gates

4 Introduction

4.1 What is thermodynamic computing?

Thermodynamic computing [1] proposes that the natural relaxation of a physical system coupled to heat baths can be harnessed to perform computation. In the quantum regime, autonomous thermal machines composed of a few interacting qubits can act as “thermodynamic neurons”: logical inputs are encoded in the temperatures of finite-size heat baths, and the logical output is read from the temperature of an auxiliary output bath once the system has settled into a nonequilibrium steady state (NESS) [2, 3]. The key appeal is that such devices operate without an external clock: computation is driven entirely by the flow of heat from hot to cold reservoirs.

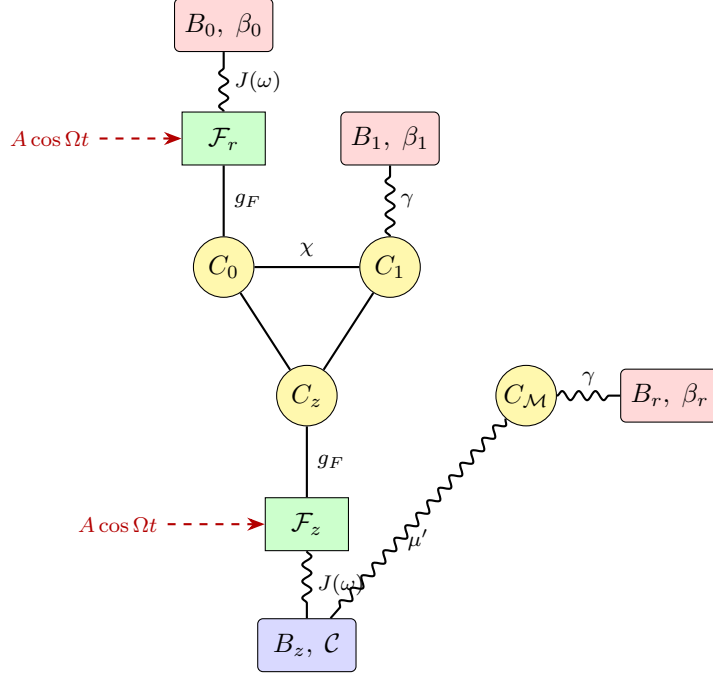


Figure 2: Floquet-buffered thermodynamic NOT gate. Yellow: machine qubits; green: driven buffer qubits with modulated gap $\epsilon_F(t) = \epsilon_F + A \cos \Omega t$ [Eq. (28)]; red: infinite baths; blue: finite output reservoir. Solid lines are coherent (excitation-exchange) couplings, wavy lines dissipative contacts, dashed red arrows the external drive. The input branch $C_1 - B_1$ is undressed so the logical encoding (27) is unchanged; the reference and output branches are filtered through $\mathcal{F}_r$ and $\mathcal{F}_z$ respectively.

4.2 Why add periodic driving?

All existing quantum thermodynamic neurons are *autonomous*, i.e., their Hamiltonians are time-independent. In this note we ask a natural question: can one improve the speed or accuracy of a thermodynamic neuron by adding a periodic driving field? This is a form of *Floquet engineering*: a periodically driven auxiliary system (the *Floquet buffer*) is inserted between the logical qubits and the physical baths. The buffer acts as a tunable effective environment whose spectral density can be shaped by the drive.

The price is twofold. First, the device is no longer autonomous; it requires a classical clock to maintain the drive and to read out the result stroboscopically. Second, an external work input is needed to power the drive. Nevertheless, the resulting architecture—a *periodically driven dissipative logic gate*—is an interesting candidate for low-energy computation.

4.3 Scope and structure of this note

This note is written as a self-contained pedagogical resource for a collaborative project involving a theoretical physicist, a mathematician working on tensor networks, and an MSc student with a background in engineering and coding. We assume familiarity with basic quantum mechanics (Hilbert spaces, operators, partial trace, von Neumann equation) and statistical mechanics (canonical ensemble, partition function). All more advanced concepts (GKSL equation, Floquet theory, chain mapping of baths, matrix-product states) are reviewed in the main text or in the appendices.

The note is organised as follows.

- Section 5 provides concise reviews of the GKSL master equation, Floquet theory, the chain representation of bosonic baths, matrix-product states (MPS), and the trace distance as

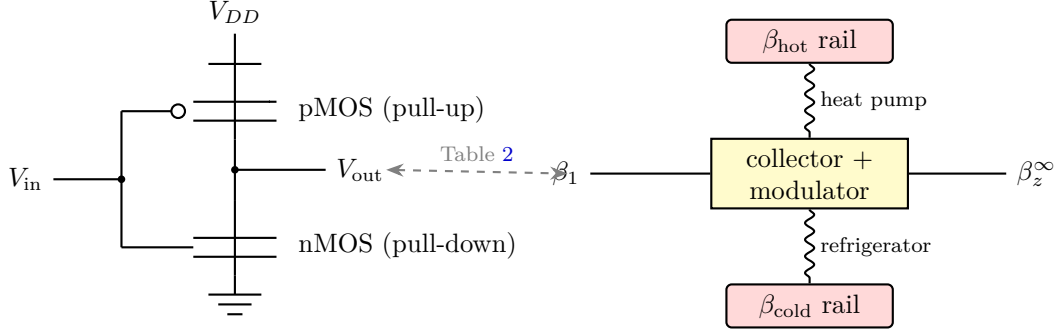


Figure 3: Structural correspondence. Left: static CMOS inverter; the complementary pair pulls the output to one rail and cuts off the path to the other. Right: the neuron’s two thermodynamic regimes play the roles of the pull-up/pull-down networks, with the modulator providing rail restoration; unlike CMOS, no regime is ever fully “cut off” — the NESS dissipates continuously, which the gated output buffer $\mathcal{F}_z$ mitigates at the cost of clock work (Sec. 3.2.4).

a distinguishability measure.

- Section 6 describes the Floquet-buffer architecture in detail.
- Section 7 states and proves a theorem (Proposition 7.1) that in a certain weak-coupling limit the logical qubits obey a time-periodic GKSL equation.
- Section 8 explains how to go beyond weak coupling using chain mapping and MPS, and how to extract energy currents and work without exponentially large bond dimensions.
- Section 9 proposes a conjectured work–distinguishability tradeoff.
- Section 10 outlines a staged numerical programme.
- The appendices contain additional technical details.

5 Background material

Here we collect, in a deliberately compact form, the mathematical tools that are essential for the rest of the note. The reader may skip this section on first reading and return as needed.

5.1 GKSL (Lindblad) master equation

A Markovian open quantum system is described by a density matrix ρ evolving under the GKSL (Gorini-Kossakowski-Sudarshan-Lindblad) equation:

$$\dot{\rho} = -i[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} \right), \quad (50)$$

where H is the effective system Hamiltonian, L_k are jump operators, and $\gamma_k \geq 0$ are positive rates. The generator is trace-preserving and completely positive. If the rates satisfy local detailed balance with respect to a heat bath at inverse temperature β , the system relaxes to a Gibbs state $\rho_\beta \propto e^{-\beta H}$ (when there is only one bath) or to a NESS when several baths are present.

5.2 Floquet theory and Floquet-Lindblad equation

A periodically driven system has a Hamiltonian $H(t) = H(t + T)$ with $T = 2\pi/\Omega$. The time-evolution operator can be written in terms of Floquet modes $|\phi_n(t)\rangle$ and quasi-energies ϵ_n :

$$U(t) = \sum_n e^{-i\epsilon_n t} |\phi_n(t)\rangle \langle \phi_n(0)|, \quad (51)$$

where $|\phi_n(t + T)\rangle = |\phi_n(t)\rangle$ and $\epsilon_n \in [-\Omega/2, \Omega/2)$.

When the system is open and weakly coupled to one or more baths, a Floquet-Lindblad equation can be derived [4, 5]. The generator becomes time-periodic:

$$\dot{\rho}(t) = \mathcal{L}(t)\rho(t), \quad \mathcal{L}(t + T) = \mathcal{L}(t). \quad (52)$$

The jump operators are expressed in the Floquet basis, and the rates depend on the Fourier components of the bath correlation functions at frequencies shifted by multiples of Ω . A crucial feature is the appearance of sidebands: transitions can occur at energies $\epsilon_m - \epsilon_n + q\Omega$, $q \in \mathbb{Z}$, which allows the system to absorb or emit quanta of the drive while exchanging energy with the baths.

5.3 Chain mapping of bosonic baths

A heat bath is usually modelled as a collection of independent harmonic oscillators:

$$H_B = \int_0^\infty d\omega \omega a_\omega^\dagger a_\omega, \quad (53)$$

coupled to the system by $H_{SB} = A \otimes \int d\omega \sqrt{J(\omega)}(a_\omega + a_\omega^\dagger)$, where $J(\omega)$ is the spectral density. For numerical purposes it is convenient to map this continuum model to a semi-infinite one-dimensional chain with nearest-neighbour hopping [6, 7]. The transformation uses a sequence of orthogonal polynomials (e.g., Lanczos or Chebyshev) determined by $J(\omega)$. After the mapping, the Hamiltonian becomes

$$H_{\text{chain}} = \sum_{k=1}^\infty \omega_k a_k^\dagger a_k + \sum_{k=1}^\infty t_k (a_k^\dagger a_{k+1} + a_{k+1}^\dagger a_k), \quad (54)$$

where a_k ($k \geq 1$) are new bosonic operators, and the system couples only to the first site: $H_{SB} \propto A \otimes (a_1 + a_1^\dagger)$. The parameters ω_k and t_k are determined recursively from the spectral density. For an Ohmic bath $J(\omega) \propto \omega$ with a suitable cutoff, the coefficients t_k decay rapidly with k , so that a truncation to a finite chain of length N (typically $N \sim 100 - 200$) gives converged results for the reduced system dynamics.

5.4 Matrix-product states (MPS) and the TDVP

A one-dimensional quantum lattice system (such as the chain-mapped bath) can be efficiently represented by a matrix-product state:

$$|\psi\rangle = \sum_{\{s_i\}} C^{s_1} A^{s_2} \dots A^{s_N} |s_1 s_2 \dots s_N\rangle, \quad (55)$$

where C and A^{s_i} are matrices, and the bond dimension χ controls the amount of entanglement that can be captured. The time-dependent variational principle (TDVP) or Trotter-Suzuki decompositions allow one to evolve the MPS in real time under a Hamiltonian $H(t)$. The computational cost scales as $O(N\chi^3 d^2)$ per time step, where d is the local Hilbert space dimension.

For the chain-mapped bath at moderate temperature, the entanglement entropy between the system and the first few chain sites saturates at a finite value. Consequently, the required bond dimension χ does not grow with the chain length N ; values of $\chi \approx 30-50$ are typical for converged simulations.

5.5 Trace distance as a distinguishability measure

The trace distance between two density matrices ρ and σ is

$$D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1, \quad (56)$$

where $\|A\|_1 = \text{Tr} \sqrt{A^\dagger A}$. It satisfies $0 \leq D(\rho, \sigma) \leq 1$ and is a metric on the space of states. Operationally, $D(\rho, \sigma)$ is equal to the maximum success probability of distinguishing ρ from σ with a single measurement. We will use it to quantify the logical distinguishability: for a binary logic gate, the two possible output states $\rho_{\text{out}}^{(0)}$ and $\rho_{\text{out}}^{(1)}$ are perfectly distinguishable when $D = 1$, and completely indistinguishable when $D = 0$.

6 Architecture of the Floquet-buffered logic gate

The physical device is sketched in Fig. 4. It consists of:

- **Logical qubits** $\mathcal{S}$, described by a time-independent Hamiltonian H_S . For concreteness, in the numerical programme we will use three qubits forming a NOR gate as in [2].
- **Floquet buffer** $\mathcal{B}_F$, a finite quantum system with a time-periodic Hamiltonian $H_{\text{buf}}(t) = H_{\text{buf}}(t + T)$, $T = 2\pi/\Omega$. The buffer could be, for example, a harmonic oscillator with a modulated frequency or a driven two-level system.
- **System-buffer coupling** $H_{S\text{buf}}$, assumed time-independent and, for the weak-coupling theorem, weak.
- **Thermal baths**, each at a different temperature T_α , coupled only to the buffer via H_{buf, B_α} .

The logical input is encoded in the temperatures $\{T_\alpha\}$ of the baths, exactly as in the autonomous thermodynamic neuron. The output, however, is not read from an auxiliary bath; instead, we measure an observable of the output qubit stroboscopically at times $t_n = nT$, after a periodic steady state has been reached. This is a deliberate departure from strict autonomy: the device requires an external clock to synchronise the drive and the readout.

The reasons for this choice are twofold. First, in the autonomous model the output bath temperature depends on the qubit-bath coupling, which is absent in our architecture (the qubits couple only to the buffer). Second, stroboscopic readout is natural in Floquet systems and is used in many quantum computing platforms.

Because we accept a clock, the thermodynamics of the device is that of a periodically driven open system. The relevant energetic quantities are the cycle-averaged work $\overline{W}$ provided by the drive and the cycle-averaged heat currents $\overline{J}_\alpha$ into the baths.

7 Weak-coupling regime: periodic GKSL generator

Under certain idealised conditions, one can eliminate the buffer and the baths and obtain a closed description of the logical qubits alone.

7.1 Assumptions

We impose:

- (A1) **Weak buffer-bath coupling.** The interactions H_{buf, B_α} are sufficiently weak that the buffer, when decoupled from $\mathcal{S}$, relaxes to a unique periodic steady state $\rho_{\text{buf}, ss}(t)$ under the Floquet-Lindblad equation. The generator of this buffer-only dynamics is $\mathcal{L}_{\text{buf}}(t)$.

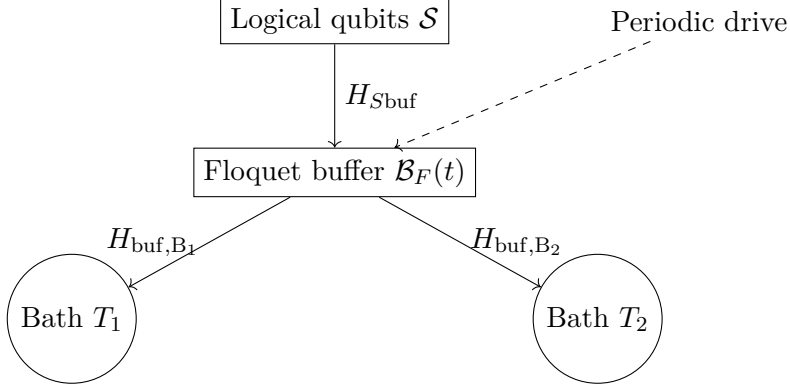


Figure 4: Periodically driven logic gate with Floquet buffer. The logical qubits interact only with the driven buffer; the buffer couples to thermal reservoirs. The device is not fully autonomous: a classical clock synchronises the drive and the stroboscopic readout.

- (A2) **Exponential decay of buffer correlations.** In that periodic steady state, the two-time correlation functions $\langle B_i(t)B_j(t-\tau) \rangle$ decay to zero on a characteristic time scale τ_c . More precisely, there exists τ_c such that $|C_{ij}(t, \tau)| < \varepsilon$ for $\tau > \tau_c$ for any chosen tolerance ε .
- (A3) **Weak system-buffer coupling and time-scale separation.** The coupling $H_{S\text{buf}}$ is much weaker than the buffer's internal relaxation scale, and the logical qubits evolve slowly compared to τ_c . This allows us to apply the Born–Markov approximation.
- (A4) **Secular approximation.** The Bohr frequencies (transition frequencies) of H_S are non-degenerate and well separated compared to the relaxation rates induced by the buffer.

7.2 Periodic GKSL generator

Proposition 7.1. *Under Assumptions (A1)–(A4), there exists a time-periodic generator $\mathcal{L}_{\text{eff}}(t)$ of the form*

$$\mathcal{L}_{\text{eff}}(t)\rho = -i[H_S + H_{LS}(t), \rho] + \sum_k \Gamma_k(t) \left(L_k(t)\rho L_k^\dagger(t) - \frac{1}{2}\{L_k^\dagger(t)L_k(t), \rho\} \right), \quad (57)$$

with $\mathcal{L}_{\text{eff}}(t+T) = \mathcal{L}_{\text{eff}}(t)$, such that the reduced state of the logical qubits obeys

$$\dot{\rho}_S(t) = \mathcal{L}_{\text{eff}}(t)\rho_S(t). \quad (58)$$

The jump operators $L_k(t)$ and rates $\Gamma_k(t)$ are obtained from the Fourier transforms of the buffer correlation functions evaluated at the Bohr frequencies of H_S , including sideband shifts by multiples of Ω .

Proof sketch. The derivation follows the standard Floquet–Born–Markov approach [4]. One starts from the Nakajima–Zwanzig equation with the buffer acting as a structured environment. The memory kernel is of second order in $H_{S\text{buf}}$. Because of (A2) the kernel decays on the time scale τ_c , and because of (A3) the system density matrix changes slowly on that scale; we can therefore make the Markov approximation and replace the memory integral by a time-local superoperator. The time dependence of this superoperator is inherited from the periodic buffer steady state. The secular approximation (A4) diagonalises the dynamics in the eigenbasis of H_S , yielding the GKSL form with time-periodic coefficients.

Practical limitations. The assumptions are strong. In particular, the weak-coupling condition (A3) may be hard to satisfy simultaneously with (A1) because weak buffer-bath coupling implies a long buffer correlation time, whereas a Markovian elimination requires the buffer correlations to decay fast *relative to the qubit dynamics*. Therefore Proposition 7.1 should be seen as identifying a parameter window in which a simple theoretical description exists, not as a statement about generic parameters.

A second issue is the dynamic Lamb shift. The Floquet buffer will induce time-dependent shifts $H_{LS}(t)$ in the qubit energy levels. The autonomous thermodynamic NOT gate relies on a precise resonance condition $\omega_W = \omega_X + \omega_Y$ among three qubit frequencies. The Floquet-induced Lamb shift will generically detune this resonance and destroy the logical operation. To mitigate this, we must actively adjust the bare qubit frequencies as functions of the drive amplitude, a procedure that is realistic in circuit-QED architectures but constitutes an additional control overhead.

For all these reasons, the main numerical programme of this project targets the regime *outside* the validity of Proposition 7.1, where no simple Lindbladian exists. Nevertheless, the weak-coupling analysis is a useful conceptual starting point.

8 Non-Markovian regime: chain mapping and MPS

To treat arbitrary coupling strengths and memory effects, we simulate the full system-buffer-baths compound as a closed many-body quantum system. We employ two established techniques: chain mapping of the baths (Sec. 5.3) and matrix-product state evolution (Sec. 5.4).

8.1 Chain-mapped Hamiltonian

Starting from the original bath model with a continuum of harmonic oscillators and a given spectral density $J(\omega)$, we apply a unitary transformation (Lanczos or orthogonal-polynomial mapping) that yields the chain Hamiltonian (54). The total Hamiltonian in the chain picture is

$$\begin{aligned} H_{\text{tot}}(t) = & H_S + H_{\text{buf}}(t) + H_{S\text{buf}} \\ & + \sum_{\alpha} \left[\sum_{k=1}^{N_{\alpha}} \omega_{\alpha k} a_{\alpha k}^{\dagger} a_{\alpha k} + \sum_{k=1}^{N_{\alpha}-1} t_{\alpha k} (a_{\alpha k}^{\dagger} a_{\alpha, k+1} + \text{h.c.}) \right] \\ & + \sum_{\alpha} V_{\text{buf}, \alpha} \otimes (a_{\alpha 1} + a_{\alpha 1}^{\dagger}). \end{aligned} \quad (59)$$

Here N_{α} is the chain length for bath α (chosen large enough to suppress finite-size effects), and $V_{\text{buf}, \alpha}$ is a buffer operator determined by the original buffer-bath coupling.

This Hamiltonian describes a one-dimensional lattice system. The logical qubits sit at the leftmost end, the buffer is next, and each bath forms a separate chain. The only interaction between qubits and baths is mediated by the buffer: the qubits couple to the buffer, and the buffer couples to the first site of each chain.

8.2 Energy current and work

Because the chain Hamiltonian is a closed system, thermodynamic observables can be defined exactly from the Heisenberg equations of motion.

The energy current from the logical qubits into the buffer is the time derivative of the qubit energy minus the contribution that would occur without the coupling:

$$J_{S \rightarrow \text{buf}}(t) = i \langle [H_S, H_{S\text{buf}}] \rangle. \quad (60)$$

Here $\langle \cdot \rangle = \text{Tr}(\rho_{\text{tot}}(t) \cdot)$, with ρ_{tot} the state of the full chain. Because H_S acts only on the qubits and $H_{S\text{buf}}$ acts on qubits and buffer, the commutator is a local operator with support on the

qubits and the buffer alone. Its expectation value can therefore be computed directly from the MPS representation of the state without needing to reconstruct the full bath density matrix.

The power injected by the periodic drive is

$$P_{\text{drive}}(t) = \left\langle \frac{\partial H_{\text{buf}}(t)}{\partial t} \right\rangle. \quad (61)$$

This is also a local observable (it acts only on the buffer). The cycle-averaged work is

$$\overline{W} = \frac{1}{T} \int_0^T P_{\text{drive}}(t) dt. \quad (62)$$

The heat dumped into bath α can be obtained from the energy current flowing from the buffer into the first site of chain α , using the buffer-chain coupling. This current is again a local observable.

8.3 Bond-dimension control

A crucial practical question is whether the MPS simulation remains tractable. For a one-dimensional chain with nearest-neighbour couplings, the entanglement entropy of the state follows an area law: for a gapped system it saturates to a constant; for a critical (gapless) system it grows logarithmically with the chain length. The chain-mapped bath is gapless (the site frequencies ω_k extend to zero), but the coupling t_k decays rapidly with k , so that only the first few tens of sites are strongly entangled with the system.

In practice, for an Ohmic spectral density with a high-frequency cutoff and for temperatures $k_B T$ above the smallest energy scale of the chain, the bond dimension required for convergence is $\chi \approx 30$ –50. This value does *not* increase when the chain length N is increased beyond the point where the first few sites have captured the relevant physics. The computational cost per time step then scales as $O(N\chi^3 d^2)$, where d is the local Hilbert space dimension. For a truncated Fock space with $d = 10$ –20, this is manageable for the single-qubit buffer prototype (Phase 3 of the numerical programme) and, with tree-tensor-network (TTN) enhancements, for the three-qubit gate (Phase 4).

9 Conjectured work–distinguishability tradeoff

In the autonomous thermodynamic neuron, logical distinguishability is intrinsically linked to the temperature difference between the baths; increasing the distinguishability requires a larger temperature gradient, which implies larger entropy production. In our driven setting, we expect a similar tradeoff: improving the logical distinguishability beyond the undriven baseline should require a positive amount of work per cycle.

Formally, let

$$P_{\text{dist}} = \frac{1}{2} \left\| \rho_{\text{out}}^{(0)} - \rho_{\text{out}}^{(1)} \right\|_1 \quad (63)$$

be the trace-distance distinguishability of the output qubit states corresponding to the two logical inputs. Let $P_{\text{dist}}^{(0)}$ be this quantity when the buffer is undriven (i.e., the baseline autonomous gate, but with the same qubit readout), and let $\overline{W}$ be the cycle-averaged work when the buffer is driven.

Conjecture 9.1 (Work–distinguishability tradeoff). *There exists a monotonically increasing function f with $f(P_{\text{dist}}^{(0)}) = 0$ such that*

$$\overline{W} \geq f(P_{\text{dist}}). \quad (64)$$

In particular, any enhancement $P_{\text{dist}} > P_{\text{dist}}^{(0)}$ requires $\overline{W} > 0$.

This conjecture is motivated by known thermodynamic uncertainty relations (TURs) and speed-limit inequalities, which bound the precision of a process in terms of the entropy production or work. The precise functional form of f and its dependence on the bath temperatures are open questions that the numerical programme will explore.

10 Numerical programme

We propose a staged numerical investigation, designed to proceed from controlled to more challenging regimes.

Phase 1: Undriven baseline. Reproduce the autonomous NOT gate of [2] using QuTiP. However, instead of reading the output from a bath temperature, we read it from the reduced state of a designated output qubit. This gives us the undriven distinguishability $P_{\text{dist}}^{(0)}$ and the baseline steady-state heat currents.

Phase 2: Weak-coupling Floquet buffer (QuTiP). Implement a single qubit coupled to a driven harmonic oscillator (the buffer), which is weakly coupled to two Ohmic baths. Use QuTiP’s Floquet-Lindblad module to numerically verify Proposition 7.1. Explore how P_{dist} and $\overline{W}$ depend on drive amplitude and frequency. Check whether the conjecture holds in this regime.

Phase 3: MPS chain simulation (single qubit). Build the chain-mapped model of one qubit, one driven buffer, and two thermal chains using ITensor or MPSDynamics.jl. Initialise the chains in thermal states (via thermofield purification). Evolve in time until the periodic steady state is reached, then compute the energy current (60), the drive power (61), and the output qubit distinguishability. Investigate parameter regimes where the weak-coupling assumptions fail, and test the conjecture beyond weak coupling.

Phase 4: Three-qubit NOR gate. Extend the chain mapping to the full three-qubit NOR gate with a single Floquet buffer. Here the bond dimension may grow; if it becomes prohibitive, we will use tree tensor networks (TTNs) or systematic truncation of the least-coupled chain sites. The goal is to produce a phase diagram of P_{dist} as a function of drive amplitude and frequency, and to test the conjecture for a practical logic gate.

All code and data will be made publicly available to facilitate reproducibility.

11 Conclusion

We have presented a self-contained, pedagogical introduction to the Floquet-buffer architecture for periodically driven logic gates. The note reviews the necessary technical background, states a rigorous theorem in the weak-coupling limit, explains how to treat the non-Markovian regime with chain-mapped MPS, and formulates a clear conjecture to be tested numerically. The project is designed to utilise the complementary strengths of the collaboration: the physicist’s formal analysis, the mathematician’s tensor-network expertise, and the student’s numerical skills. This note is the definitive starting point for the work.

A Derivation of the GKSL equation from a microscopic model

For completeness we sketch the derivation of the GKSL equation (50) from a system-bath Hamiltonian. Consider a system S coupled to a bath B via $H_{SB} = \sum_i S_i \otimes B_i$. In the Born–Markov approximation, after tracing out the bath, one obtains the Redfield equation. The additional

secular (rotating-wave) approximation diagonalises the dynamics in the eigenbasis of H_S and ensures complete positivity. The resulting jump operators are the eigenoperators $A(\omega)$ of the system Hamiltonian, with rates $\gamma(\omega) = \int e^{i\omega t} \langle B^\dagger(t) B(0) \rangle dt$. The detailed balance condition $\gamma(-\omega) = e^{-\beta\omega} \gamma(\omega)$ follows from the KMS condition of the bath. See [9] for a textbook treatment.

B Chain mapping: recurrence relations

Given a spectral density $J(\omega)$, the chain mapping is constructed by orthogonalising the polynomials in ω with respect to the measure $J(\omega)d\omega$. The recurrence coefficients ω_k and t_k are obtained from the continued fraction expansion of the bath correlation function. For the common case of an Ohmic bath with exponential cutoff, $J(\omega) = \eta\omega e^{-\omega/\omega_c}$, the coefficients can be computed analytically or numerically using the Lanczos algorithm [6].

C MPS representation of a thermal state

To initialise the chain in a thermal state at temperature T , one uses the thermofield (purification) approach. Each physical bosonic site is doubled into an ancilla site, and the initial state is the vacuum for a set of Bogoliubov-transformed modes. The resulting MPS bond dimension is directly related to the effective temperature; for moderate T , it remains small.

D Energy current operator: explicit form

Equation (60) follows from the Heisenberg equation for H_S :

$$\frac{dH_S}{dt} = i[H_{\text{tot}}, H_S] = i[H_S, H_{S\text{buf}}] + i[H_S, H_S] = i[H_S, H_{S\text{buf}}]. \quad (65)$$

The term $[H_S, H_S] = 0$, and H_S commutes with all bath and buffer-bath Hamiltonians because it acts on a different tensor factor. Hence the energy change of the qubits is solely due to the qubit-buffer interaction. Taking the expectation value gives the current.

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